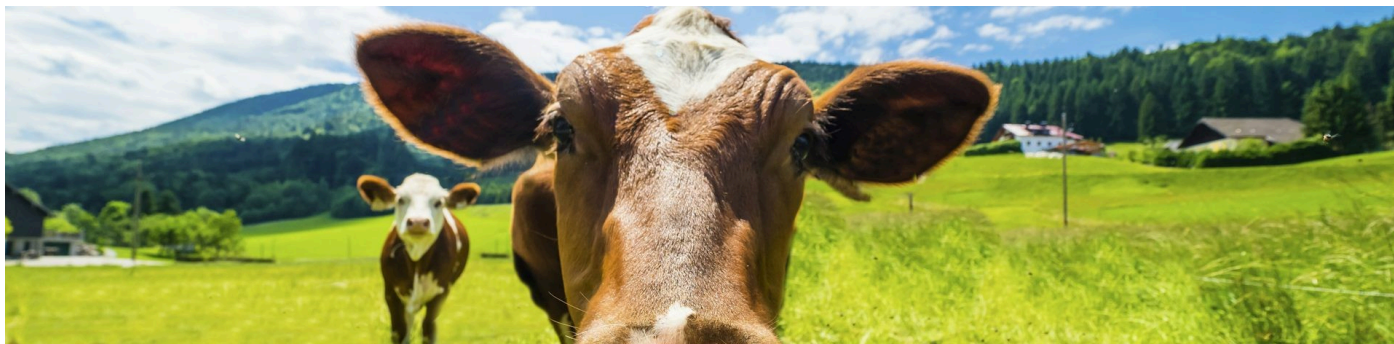


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Unpacking Methane: Livestock, Fossil Fuels, and Climate Policy

Abstract

In 2023, New Zealand proposed taxing the methane that cows burp. Climate scientist Kevin Trenberth asked whether methane from cows is really as harmful as methane from fossil fuels. His answer depends on where the carbon comes from and how long it lasts, and it has big implications for how countries design climate policies to reach net zero.

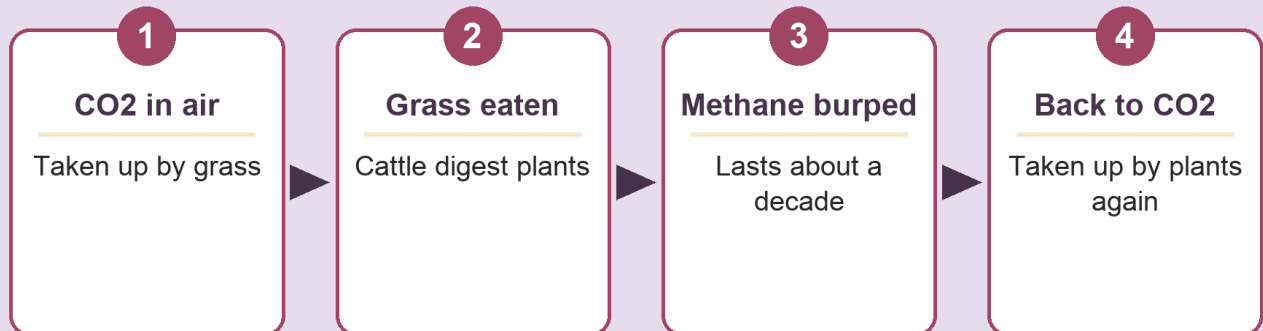
Introduction

In New Zealand, agriculture is one of the largest contributors to climate change, and the country's warming effect comes roughly equally from carbon dioxide and methane. With a goal of net-zero emissions by midcentury, it proposed a tax on livestock methane. Worldwide, 150 governments have committed to cutting methane, both from agriculture and from the largest source, leaks from fossil fuel infrastructure. Methane is the second most important greenhouse gas. It is many times more potent than carbon dioxide at warming the climate, but it lasts only about a decade in the atmosphere. So is all methane equally bad, and how should it be counted?

How Do Scientists Study This?

Trenberth compared methane and carbon dioxide by looking at both their potency and lifetimes. Policymakers commonly convert methane to a carbon dioxide equivalent using a global warming potential of 28 over 100 years. Trenberth examined how that single number misrepresents methane: a methane spike has no effect after about 30 years, so its impact is overstated over centuries and understated over the first 20 years. He then traced the carbon cycle behind two sources. Biogenic methane from livestock starts as carbon dioxide taken up by grass. Fossil methane comes from carbon stored underground for millions of years. Comparing these cycles shows their different long-term effects.

The biogenic methane cycle



Compare lifetimes

Methane lasts about a decade; some carbon dioxide lasts thousands of years.

Question the conversion

One global warming potential number hides very different short- and long-term effects.

Trace the carbon

Livestock methane recycles recent carbon; fossil methane adds ancient carbon.

What Did They Discover?

Biogenic methane has a circular life: grass takes up carbon dioxide, cattle eat grass and burp methane, and after about a decade the methane becomes carbon dioxide that plants take up again. Livestock numbers, not the cycle itself, drive increases, as demand for animal protein grows. Fossil methane, by contrast, adds new carbon to the atmosphere. Livestock produce about one-third of human-caused methane, while oil and gas operations represent about 63%. Leaks can be huge: the 2022 Nord Stream pipeline leak released a reported 500,000 tons of methane. Reducing methane buys about a decade of lower temperatures, while reducing fossil carbon buys thousands of years. Tree offsets are also temporary and often overestimated.

Methane by the numbers

~10 yrs

methane lasts in the atmosphere

63%

of human methane from oil and gas operations

500,000 t

of methane reportedly leaked from Nord Stream

Trenberth argues fossil fuel emissions can only be credibly offset by storing carbon dioxide in a form stable for many thousands of years.



FROM THE ARTICLE

Methane leaks visible in 2019 satellite data from the Permian Basin in Texas and New Mexico. Image: Global Airborne Observatory/Carbon Mapper, University of Arizona/Arizona State University/NASA/JPL-Caltech.

Takeaways

Not all methane is equal

Livestock methane recycles recent carbon; fossil methane adds new carbon.

Time frames matter

One conversion number hides methane's short, intense warming effect.

Fossil fuels first

Cutting fossil fuel use is the lasting solution to climate change.

Key Terms

Biogenic methane

Methane produced by living things, such as livestock digestion and decaying manure.

Global warming potential (GWP)

A number comparing how much heat a gas traps relative to carbon dioxide over a set time period.

Circular economy

A system in which resources are reused and recycled rather than used once and discarded.

Carbon offset

An action, such as planting trees, meant to balance emissions released elsewhere.

Short-lived climate pollutant

A greenhouse gas or particle that stays in the atmosphere for a short time but has a strong warming effect.

Activity

Two Carbon Cycles

Students diagram the biogenic and fossil carbon cycles and evaluate a methane tax policy.

Materials: Poster paper; Arrow cutouts or markers; Policy evaluation worksheet

1. Draw the biogenic cycle: atmosphere CO₂ → grass → cow → methane → CO₂ after about a decade.
2. Draw the fossil cycle: underground carbon → extraction and leaks → methane and CO₂ added to the atmosphere.
3. Label which cycle adds new carbon to the atmosphere and which recycles it.
4. Evaluate New Zealand's cow burp tax: list one argument for it and one concern raised by Trenberth.
5. Propose a policy that treats short-lived and long-lived gases differently.

References

Trenberth, K. (2023). New Zealand wants to tax cow burps: here's why that's not the best climate solution. The Conversation. <https://doi.org/10.64628/AAI.fmn6wyant>